

# The question

We only ever used **100** of FoldBench's **334** monomers. #213 showed the contact eval set is 58% homologous to exp199's 70.9M training sequences, and that what survives a homology filter is mostly de novo *designs* — the **natural** count bottoms out around 50, too thin to measure novel-protein performance rather than bound it.

So: if we grow the eval set with the FoldBench monomers we never used, how many of the expanded set survive a sequence-identity filter against our training data, at 40% and at 30%?

No new inference. exp213's 17 GB MMseqs2 target database is reused as-is; this appends 222 net-new queries to its 554 and re-runs the search. About seven minutes on the workstation.

# The set, verified not assumed

FoldBench's `monomer\_protein.csv` at exp12's pinned commit `4273f687`, sha256-verified, has 334 rows, and our 100 are exactly its first 100 — checked rather than taken on faith. That leaves 234 unused. **\*\*Twelve of those are already eval proteins\*\*** under exp65's `denovo\_pdb` label and would have been double-counted, so the real increment is **\*\*222 net-new\*\*** and the expanded set is **\*\*776\*\***.

Sequences come from RCSB's canonical `entity\_poly` field. FoldBench's `chain\_id` is sometimes the mmCIF *label* asym id rather than the auth chain — ten entries, not the five the issue listed — so chains resolve auth-first with a label fallback and fail loudly on ambiguity.

Both residue checksums land exactly (66,692 aa and 64,624 aa), and all 100 of the sequences we already use are reproduced **\*\*byte-for-byte\*\*** through the new fetch path.

# Parity with #213 is exact

Re-running exp213's 554 queries through the expanded search reproduces its **\*\*284 and 264\*\*** survivors at the two thresholds, and its 273 and 255 without the coverage gate, with **\*\*zero of the 554 rows\*\*** changing identity or stratum. The expanded table is a strict superset of exp213's and joins to it on dataset and stem.

One correction to the plan on the issue: its quoted command line says `--max-seqs 2000`, but exp213's published table was built with **\*\*5000\*\*** — 2000 was its AFDB-only cross-check. Parity means matching the published table.

# Result — 23 survivors, not the 33 predicted

Of the 222 net-new proteins, **23** survive a <40% identity filter and 11 survive <30%, taking the expanded eval set to 307 of 776 and 275 of 776.

**The extrapolation on the issue was optimistic at both thresholds**, which is the question it could not answer for itself. It predicted 33 and 24. The 100 monomers we already use are the *oldest-deposited* rows of a PDB-ID-sorted file, and the newer entries turn out to be **more** homologous to our training data, not less: 10.4% survive at <40% against 15.0%, and 5.0% at <30% against 11.0%. Neither gap clears  $p < 0.05$  alone (Fisher 0.26 and 0.057), so this reads as a consistent shortfall rather than a proven rate difference — but the counts the eval set actually gets are 23 and 11.

Length is not the explanation: median 242 aa against 247.5 aa.

# The natural count is what decides this

Every one of the 23 new survivors is a **natural** protein. The decontaminated natural count goes from **55** to 78 at **<40%**, a gain of 23 or **+42%**, and from **50** to 61 at **<30%**, a gain of 11 or **+22%**.

The baseline is **55**, not the 58 the issue quotes. #213 splits designed from natural on the dataset label, which cannot see a designed protein sitting in a FoldBench row — and one can, since 12 FoldBench monomers are themselves in exp65's de novo set. Resolving each FoldBench entity's RCSB source organism finds that 3 of #213's 15 FoldBench-100 survivors at **<40%** are synthetic constructs. The proxy is calibrated, flagging 12 of 12 known designs, and deliberately conservative, since it also catches engineered variants of natural proteins. Only 4 of the 222 trip it, and none of those 4 survive either filter.

# Both arms, and why one is not enough

exp199 trained on **both** corpora, so the union is the filter that counts — 4.13M AFDB sequences with AlphaFold2 labels plus 66.76M ESM-Atlas sequences with ESMFold2 distillation labels. Every number above uses the union. But every prior overlap check (#41, #65, #94) only ever looked at AFDB, and that turns out to matter a great deal.

Against **AFDB alone**, 76 of the 222 net-new monomers would look clean at <40%. Against **ESM-Atlas alone**, 62. Against **both**, only **23**. A single-arm check overcounts the clean set by about 3x, and the two arms are largely complementary rather than redundant: ESM-Atlas removes 53 proteins AFDB alone would have kept, and AFDB removes 39 ESM-Atlas alone would have kept.

**The pattern reverses between the two FoldBench slices.** Of the 199 net-new dropped at <40%, 107 are reachable from both arms, 39 from AFDB alone and **53 from ESM-Atlas alone** — the metagenomic half is the larger sole contaminator. For the existing 554 the same computation gives 183 / 60 / 27, where AFDB dominates, and reproduces the figures on the issue exactly — a third independent parity check. The ESMFold2-distillation corpus is doing *more* of the contaminating on newer PDB entries, which is part of why extrapolating from the older 100 came out optimistic.

# eval2 — the filtered set

`data/eval2\_manifest.csv` is the deliverable that follows from all of this: the expanded set with **every protein at or above 40% training identity removed**, leaving **307**, sequences included and annotated so a stricter cut costs no new compute.

`best\_identity` is the coverage-gated maximum over **both** arms, so `best\_identity < 0.30` reproduces the 30% set (**275** proteins) exactly; `passes\_30` is precomputed. Per-arm columns allow the same cut against AFDB or ESM-Atlas alone, and `best\_identity\_ungated` is the paranoid bound — 18 of the 307 clear 40% only because of the 50% coverage gate.

One property constrains what eval2 can measure, and it is a column rather than a caveat in prose. **75% of it (229 of 307) is de novo designed protein** — not a choice made here, but what survives a homology filter, and exactly the confound #213 raised; `designed\_any` splits it and the natural subset is **78** at 40% and **61** at 30%.

# Ground truth for all 307

23 of eval2's proteins were outside #89's frozen GT universe and so unscorable. Their contacts are now computed with **#89's own `compute\_contacts`** — imported, not reimplemented — on the RCSB `-assembly1`` mmCIFs exp12 used for the FoldBench-100, emitted in #89's exact schema. The two files concatenate into a **577-unit** universe. All 23 come out clean: alignment identity **1.000** for every one, resolved/L 0.83–1.00, 202–1046 contacts each.

**The control is what makes them usable.** Running the **new** code path on the 100 FoldBench proteins #89 already published reproduces **100/100 records exactly** — L, n\_resolved, chain, alignment identity, the resolved set and every (i, j, degree) contact. That includes all six label-chain entries, where #89 passed FoldBench's label id and fell back to the longest polymer chain, landing on the same auth chain this now passes explicitly. So the 23 are scored on the same definition of "contact" as the 554 and can be pooled with them.

Published to the bucket at ``data/contacts-v1-eval2-exp226/``; nothing under #89's prefix was touched.

# The scoreboard — six predictors, 307 proteins

Every comparator was run on the 23 new proteins and joined to the 284 already scored, giving the first complete measurement on a decontaminated set. R-precision (all ranges):

pooled eval2 (307) — Marifold **0.545**, Protenix single-seq 0.679, ESMFold 0.694, ESMFold2 0.740, Protenix+MSA 0.778, seq-KNN 0.074. On that pooled number Marifold is last among the real predictors.

But pooled eval2 is **75% de novo design**. On the 78 **natural** proteins the ordering changes: Marifold **0.337** against Protenix single-seq **0.326**.

# The Protenix parity comes back

#213 concluded the parity with Protenix-v2 single-seq "does not survive" homology removal — it lost by 0.169 there. **\*\*That subset was ~80% de novo designs.\*\*** Paired bootstrap, 10,000 resamples:

eval2 natural (78) **\*\*+0.011\*\*** [-0.044, +0.069] — a tie. eval2 natural at <30% (61), -0.041 [-0.101, +0.019] — also a tie. The 23 net-new (23), **\*\*+0.164\*\*** [+0.063, +0.263] — MarFold wins outright. Pooled (307), -0.133 — Protenix wins.

So #213's finding was a **\*\*composition effect, not a homology effect\*\***. Protenix single-seq is unusually strong on idealised de novo backbones and unusually weak on novel natural proteins: it collapses to **\*\*0.243\*\*** on the 23 (against 0.603 on the full 554) while its MSA mode reaches **\*\*0.805\*\*** — the widest single-seq/MSA gap in the experiment. What that comparison turned on was removing *\*designs\**, not removing *\*homologs\**.

**\*\*The tie is a cancellation, not a match.\*\*** Protein by protein the two agree only weakly (Pearson  $r=0.40$ ); the mean absolute difference is 0.192, seventeen times the +0.011 mean. MarFold alone clears 0.5 on 8 proteins, Protenix alone on 13, both on only 5. They are strong on largely disjoint sets, which makes an ensemble the obvious follow-up.

**\*\*What has not changed:\*\*** MarFold still loses to ESMFold (-0.125), ESMFold2 (-0.192) and Protenix+MSA (-0.361) on eval2-natural, all significant. This does not close the gap to the structure predictors.

**\*\*The null collapses as designed:\*\*** seq-KNN falls to 0.074 pooled and 0.148 natural, with MarFold +0.189 [+0.141, +0.236] over it. The decontamination did what it was meant to do and the score is not retrieval.

# Verdict

**\*\*Fold it in at <40%; it changes little at <30%.\*\*** A +42% increase in decontaminated natural proteins is a real gain on exactly the axis #213 said bottoms out. The +22% at <30% is not enough to change what the eval set can measure.

FoldBench is confirmed as the dirtiest slice we have. **\*\*89.6%\*\*** of the net-new monomers fail a 40% filter, and **\*\*all 222\*\*** have at least some alignment into the training set, so the expansion adds only **\*\*8\*\*** proteins to #213's "no detectable homolog" stratum, 231 to 239.

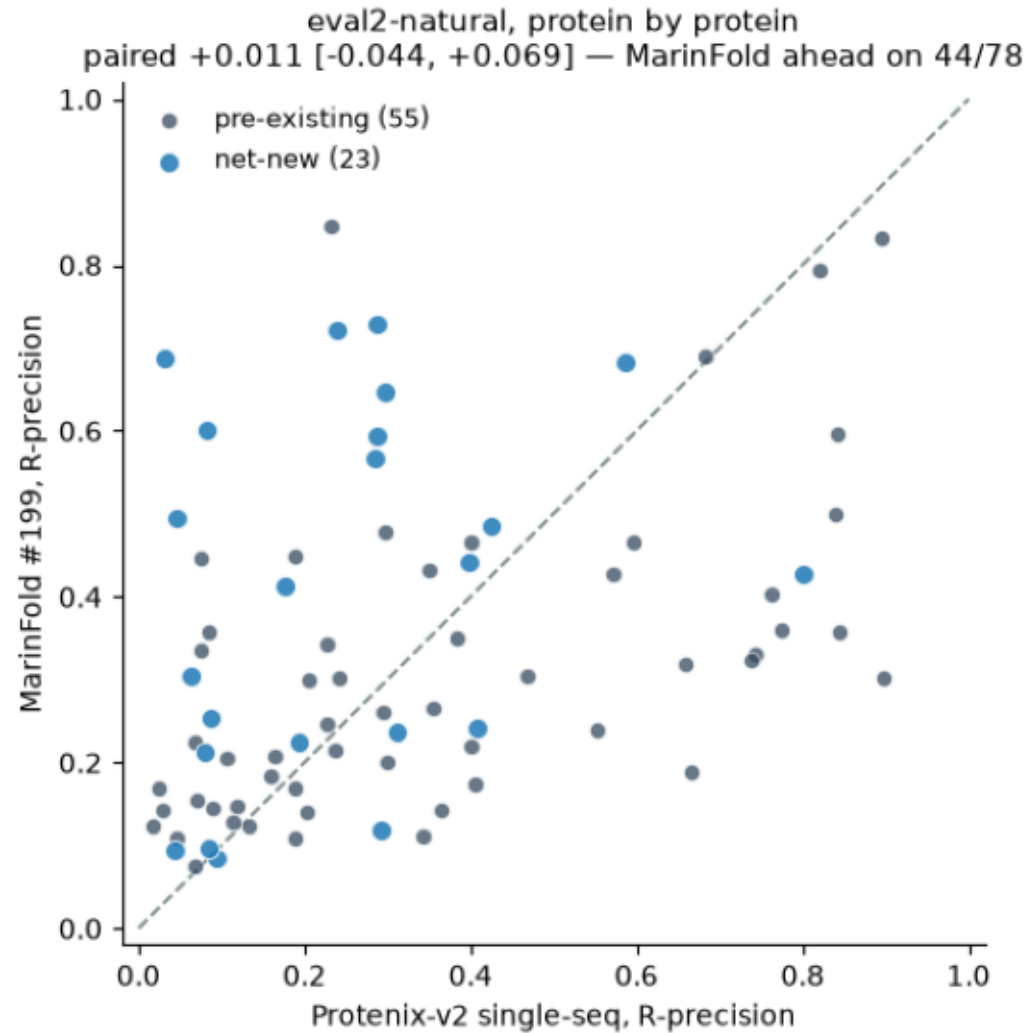
Recommendation: add `foldbench\_rest` as its own stratum rather than merging it into `foldbench100`, since the two have measurably different training-set proximity and any future "first N FoldBench rows" would inherit the same deposition-date bias.

**\*\*Read eval2-natural, not eval2 pooled.\*\*** The pooled number is 75% designed and mostly reports how well each model folds idealised backbones; it is the reason #213's conclusion needed revising. n=78 is still thin, so treat the tie as a tie rather than a result in either direction.

Not measured here: the **\*\*fold-novel\*\*** count for the 222, which needs a Foldseek pass against exp41's Modal-hosted AFDB representative DB — beyond this issue's sequence-search budget. The obvious follow-ups are that Foldseek axis, and a MarinerFold + Protenix-single-seq ensemble, which the  $r=0.40$  per-protein disagreement says should beat either alone.

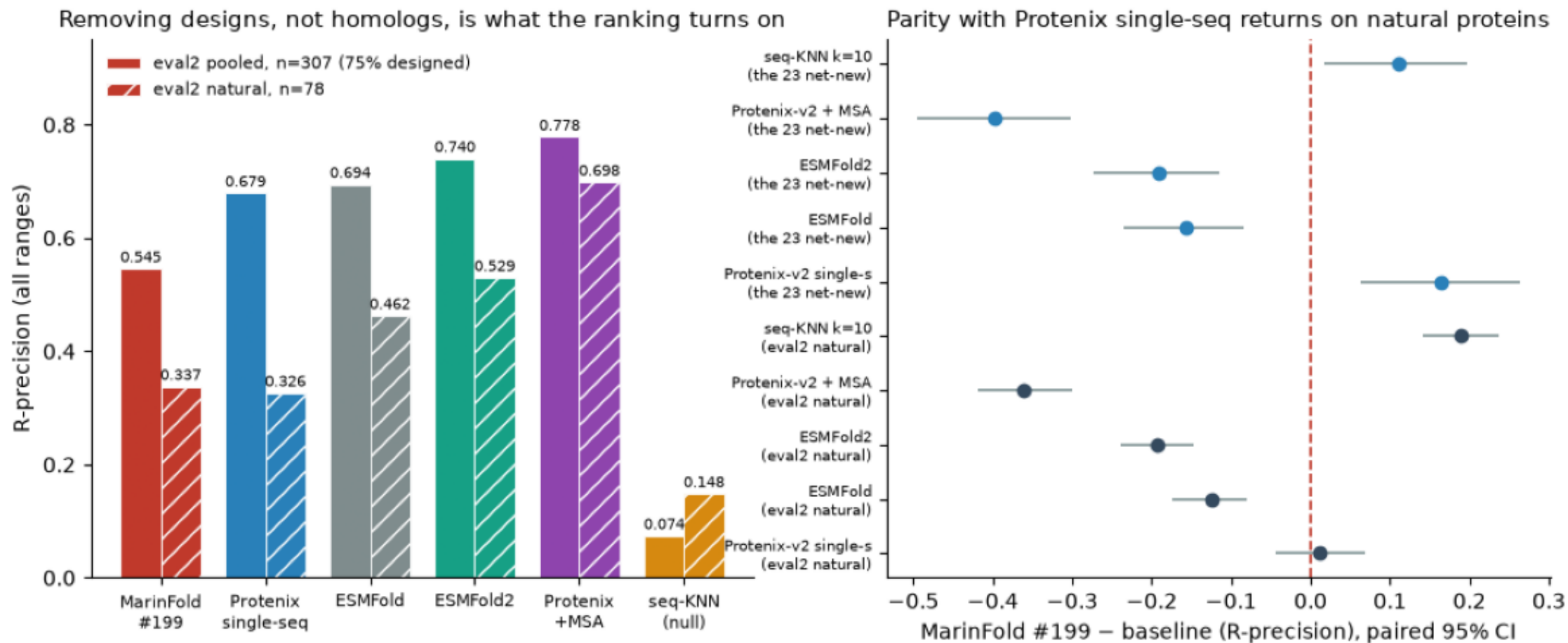
## eval2\_natural\_head\_to\_head.png

Each point is one natural eval2 protein. The tie is a cancellation, not a per-protein match: they agree only weakly ( $r=0.40$ ) and the mean absolute difference is 0.192, 17x the +0.011 mean.



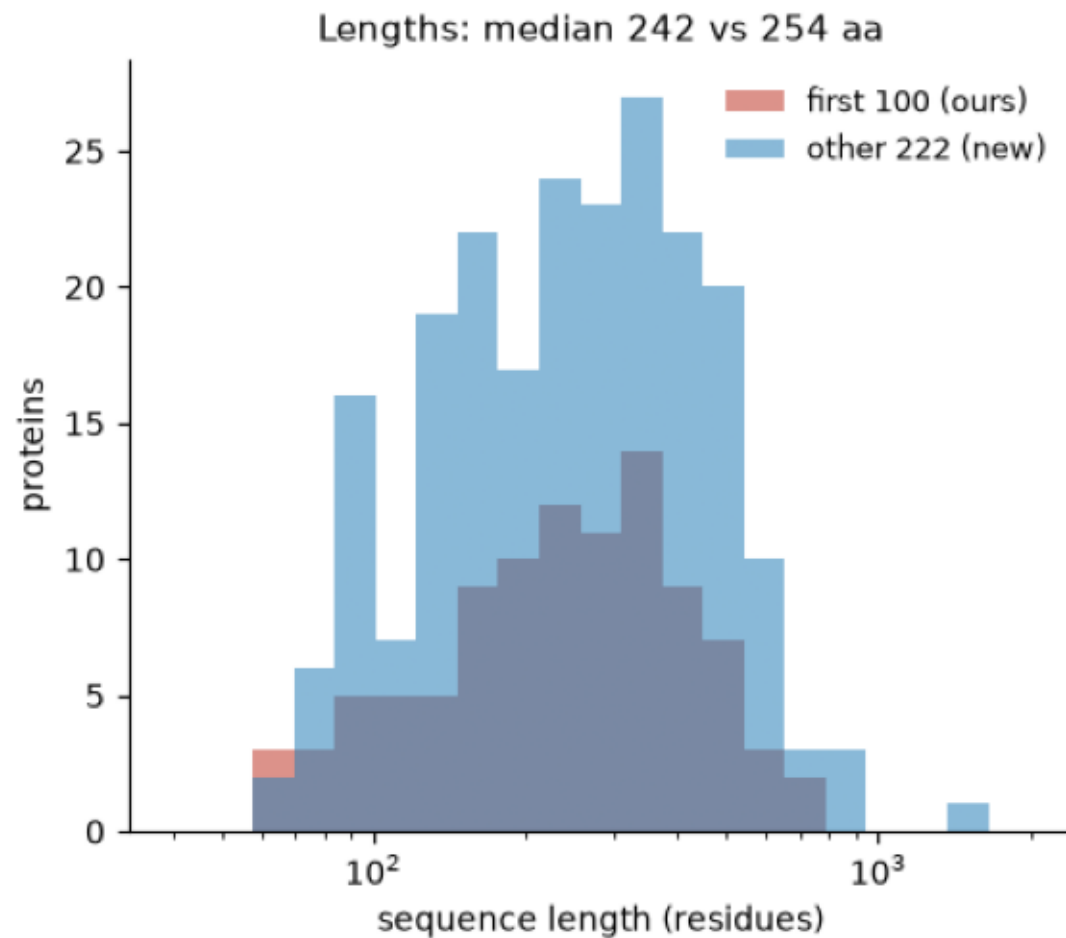
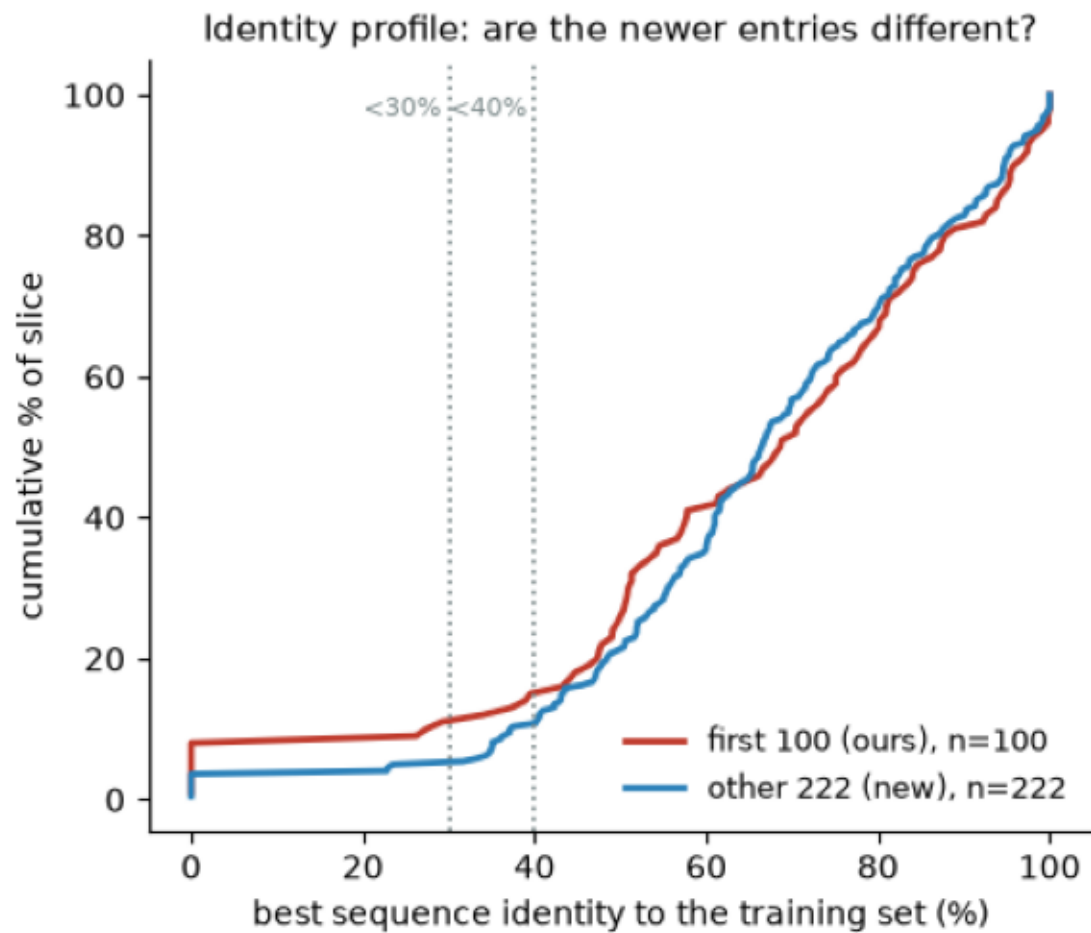
## eval2\_scoreboard.png

Left: eval2 pooled is 75% de novo design, and every structure predictor beats MarinFold there. Restricted to the 78 natural proteins the ordering changes and MarinFold matches Protenix-v2 single-seq. Right: paired MarinFold-minus-baseline differences with 10,000-resample bootstrap CIs; intervals crossing the dashed line are ties.



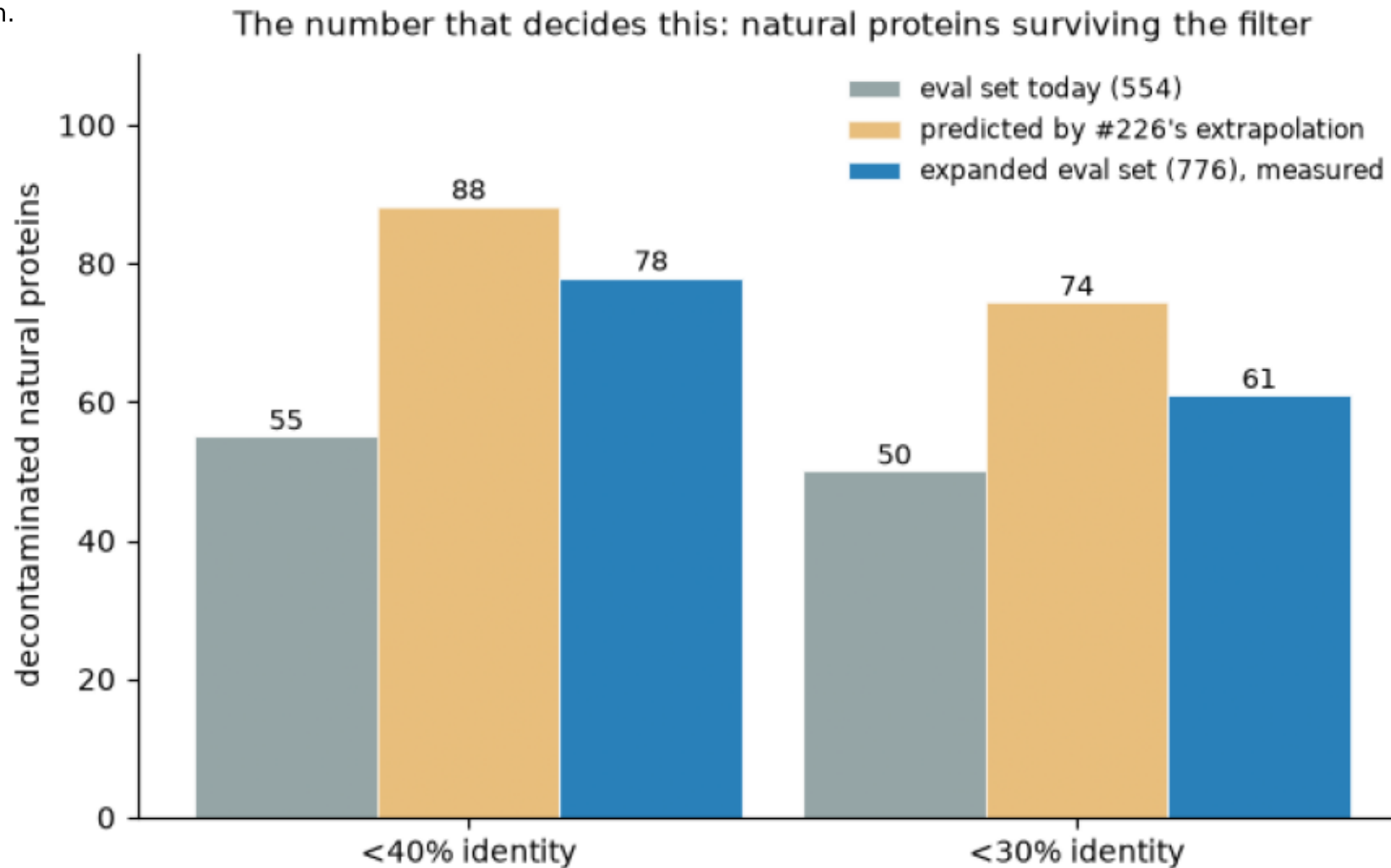
## identity\_profile\_old\_vs\_new.png

Left: cumulative distribution of best training-set identity; proteins with no covered hit are plotted at 0. The new curve sits to the right of the old one below 40 %, which is the shortfall against the extrapolation. Right: the length profiles match, so length is not the explanation.



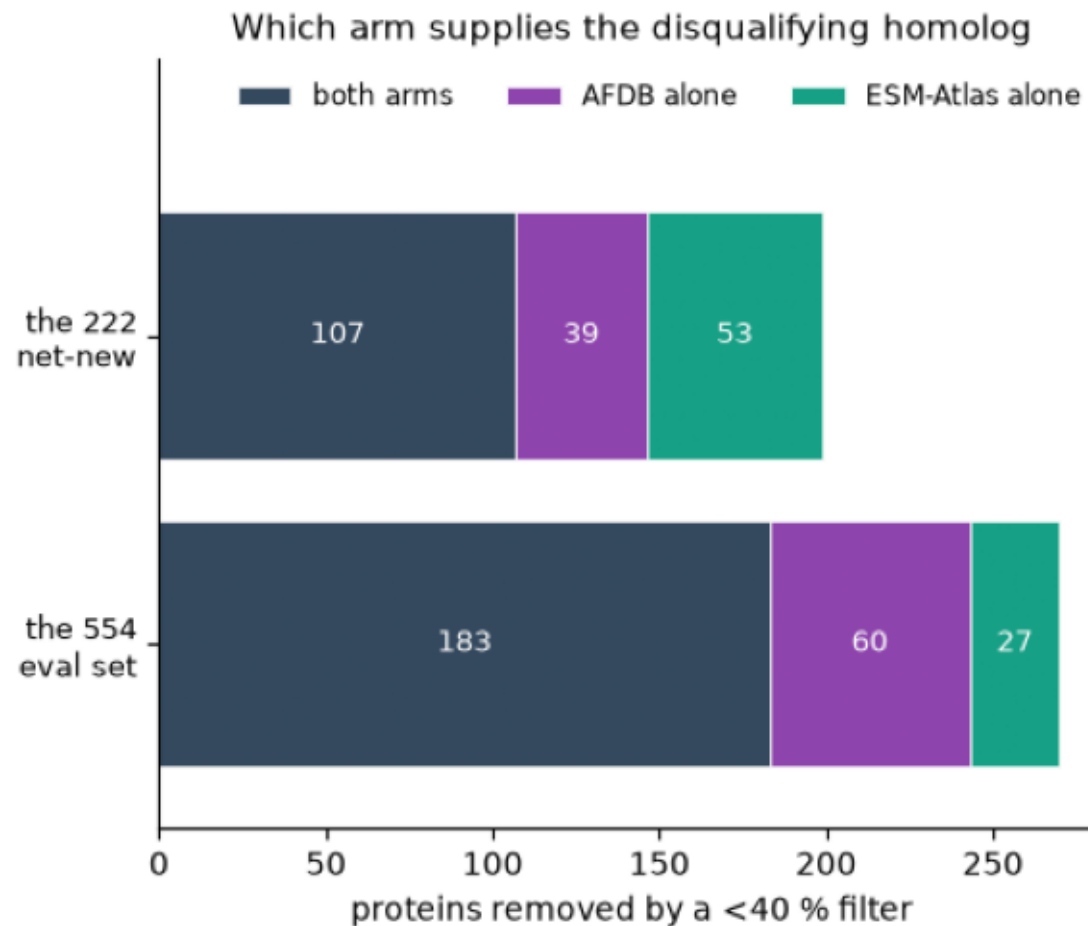
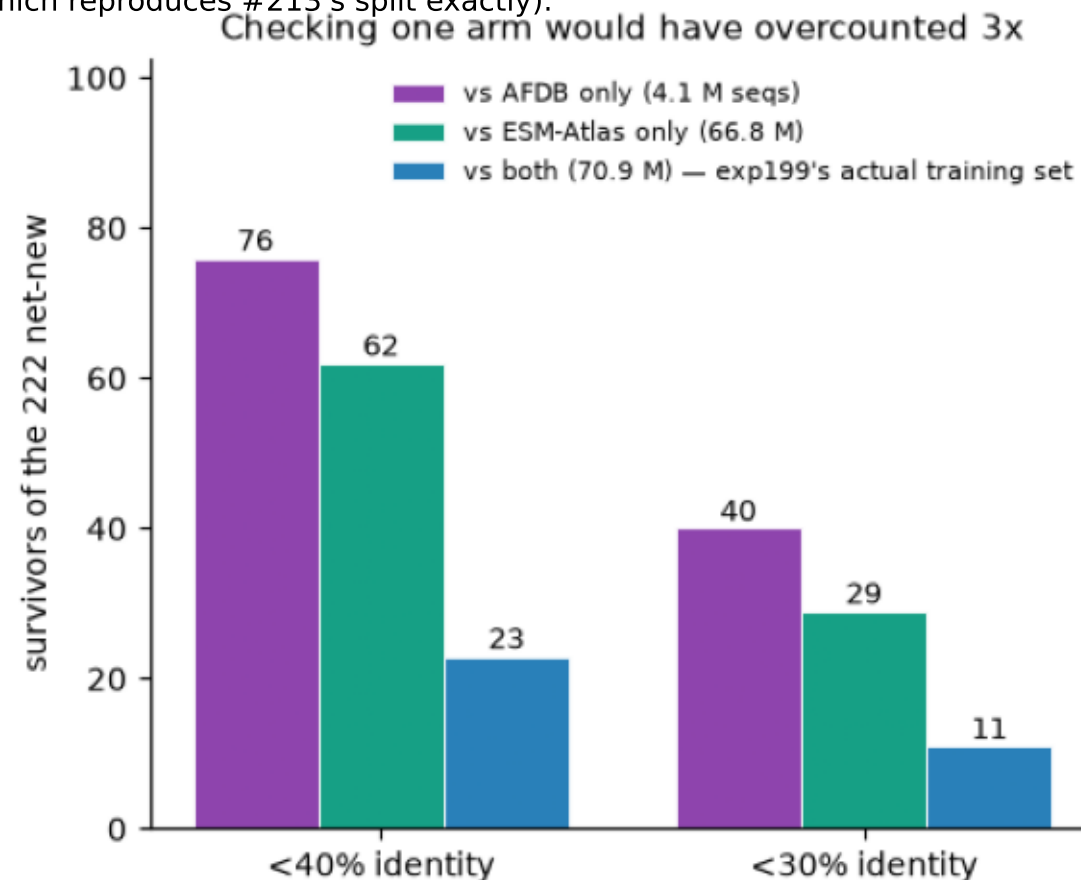
## natural\_gain.png

Natural (non-designed) eval proteins with no training homolog above the threshold. 'Natural' excludes both exp65's de novo set and any FoldBench entity whose RCSB source organism is synthetic. The expansion delivers its <40 % gain but only about half the predicted <30 % gain.



## per\_arm\_survival.png

exp199 trained on both corpora, so the union is the filter that counts — but every prior overlap check (#41, #65, #94) looked at AFDB only. Left: AFDB alone would have left 76 of the 222 looking clean at <40 %; the union leaves 23. Right: for the net-new set the metagenomic ESM-Atlas arm is the \*larger\* sole contaminator (53 vs 39), reversing the pattern in the existing 554 (27 vs 60, which reproduces #213's split exactly).



## survival\_by\_dataset.png

Share of each eval-set slice with no training sequence above the identity threshold (MMseqs2 -s 7.5, hit counted at  $E \leq 1e-3$  and  $qcov \geq 0.50$ ). Bar labels are survivors/total. The two FoldBench slices are the dirtiest in the eval set; the 222 we never used are dirtier still than the 100 we do.

